

Auteur: Thijs Van Schel
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 42.18

$$\int \frac{2x^2 - 5x + 7}{x^3 - 7x + 6} dx$$

$$1 \left| \begin{array}{ccc|c} 1 & 0 & -7 & 6 \\ & 1 & 1 & -6 \\ & 1 & 1 & -6 \\ & 1 & 1 & -6 \end{array} \right| \begin{array}{c} 6 \\ -6 \\ 0 \\ 0 \end{array}$$

$$\Rightarrow (x-1)(x^2+x-6) = (x-1)(x-2)(x+3)$$

$$\int \frac{2x^2 - 5x + 7}{(x-1)(x-2)(x+3)} dx$$

$$\frac{2x^2 - 5x + 7}{(x-1)(x-2)(x+3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$$

$$A = \left. \frac{2x^2 - 5x + 7}{(x-2)(x+3)} \right|_{x=1} = -1$$

$$B = \left. \frac{2x^2 - 5x + 7}{(x-1)(x+3)} \right|_{x=2} = 1$$

$$C = \left. \frac{2x^2 - 5x + 7}{(x-1)(x-2)} \right|_{x=-3} = 2$$

$$\Rightarrow - \int \frac{1}{(x-1)} dx + \int \frac{1}{(x-2)} dx + 2 \int \frac{1}{(x+3)} dx$$

$$= -\ln|x-1| + \ln|x-2| + 2\ln|x+3| + C$$

References